

PROJECT OVERVIEW DOCUMENT

ChopWise

Smart Restaurant Reservation & AI Recommendation Platform
for Douala, Cameroon

Document Purpose: Complete system overview for non-technical readers

City of focus: Douala, Cameroon · **Currency:** FCFA (XAF)

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1. The Problem We Are Solving in Douala

Douala is the largest city and the economic capital of Cameroon. It hosts thousands of restaurants — from neighbourhood *tournedos* spots in Akwa and Bonanjo, to family restaurants in Bonapriso, fish grills in Deïdo, and upscale dining in Bonamoussadi. Yet, when a resident or a visitor wants to find a place to eat, plan a celebration, or simply book a table for the evening, the entire experience is still informal, fragmented, and frustrating.

1.1 The Specific Pain Points in the City

No central place to discover restaurants

Diners rely on word of mouth, Facebook pages, or WhatsApp status. There is no single reliable directory that shows menus, prices, opening hours, and locations together.

Reservations happen by phone — or not at all

To "book" a table, customers call the restaurant. Many calls go unanswered, lines are busy, or the staff forgets to write the booking down. Walk-ins are turned away on busy nights.

Hidden prices, surprise bills

Without a public menu, diners do not know what a meal will cost until they sit down. This blocks people on a budget from confidently trying new places.

Restaurants lose customers they never knew they had

Owners cannot reach diners online, cannot show their dishes, and have no way to track how many bookings they receive each week.

No-shows hurt small restaurants

Without a commitment from the diner, restaurants prepare tables and food for guests who never arrive — a direct loss for businesses with thin margins.

No digital presence for most local restaurants

Many great Cameroonian restaurants have no website at all. They are invisible to tourists, expatriates, and young diners who search online first.

No AI to help diners decide based on budget

Even global tools like Google Maps and Google My Business only let you search and filter. They cannot answer a real human question like "I have 2,500 FCFA, what can I eat tonight near me?" Diners are left to scroll and guess.

The core problem in one sentence: In Douala, diners and restaurants are not properly connected. Discovery is informal, reservations are unreliable, prices are hidden, and small restaurants have no digital tools to grow. ChopWise solves this gap.

1.2 The Missing Piece: An AI That Recommends by Budget

The biggest gap in the dining experience — not only in Douala, but globally — is that **no major platform actually helps a diner decide what to eat based on what they can afford right now**. Search tools like Google Maps, Google My Business, TripAdvisor, and even local Facebook groups all do the same thing: they show a list and leave the diner to scroll, compare, and guess. They expect the user to already know the answer.

But this is not how people in Douala actually make dining decisions. A student walking home from class, a worker after a long day, or a couple on a Friday evening usually starts with a budget in mind first: *"I have 2,000 FCFA", "We have 10,000 FCFA for two", "I want something under 5,000 FCFA that delivers in Bonapriso."* No existing platform answers that question directly.

ChopWise bridges this gap with a built-in **AI Recommendation Engine**. A diner can simply tell the AI their budget, their location, what kind of food they feel like, or even a special occasion, and the AI returns a short, sensible list of restaurants and even specific dishes that fit. The recommendations are grounded in real menu data from real restaurants on the platform — not generic search results — which means every suggestion is something the diner can actually order and book right away.

Why this is unique: Google Maps tells you *where* restaurants are. Google My Business tells you *what hours* they keep. ChopWise tells you *"given your 3,000 FCFA budget tonight, here are 4 restaurants near you, and here is what to order at each one to stay in budget."* That is a level of personal assistance no global mapping platform offers, and it is exactly what diners in Cameroon need.

1.3 Why This Matters Now

Douala's middle class is growing, smartphone use is widespread, mobile money (MTN MoMo, Orange Money) is mainstream, and people increasingly expect the same digital convenience they see in services like Uber, Yango, or Jumia. AI assistants like ChatGPT have also taught users to *ask* for help rather than *search* for it. Yet the restaurant sector — and the broader local discovery sector — has been left behind. ChopWise brings both the convenience of online booking **and** the intelligence of a personal AI dining assistant to Cameroon.

2. What ChopWise Is

ChopWise is a web platform — accessible from any phone or computer — that lets people in Douala (and across Cameroon) **discover restaurants, view real menus with real prices in FCFA, reserve a table online, and confirm the booking with a small fee paid by mobile money.** Restaurants get a digital profile, a way to manage their menu and bookings, and a steady stream of confirmed customers.

In short: ChopWise is to restaurants in Cameroon what online booking platforms are to travel — a trusted middleman that gives diners confidence, gives restaurants visibility, and protects both sides through a clear booking and payment flow.

2.1 The Three Promises of the Platform

- **For diners:** See the menu and the price before you go. Ask an AI for a restaurant that fits your budget. Reserve a table from your phone. Know your reservation is confirmed.
- **For restaurants:** Be found online and recommended by AI to the right diners. Receive serious bookings only. Reduce no-shows. Manage your menu without paperwork.
- **For the city:** A modern, AI-assisted dining ecosystem that supports local restaurants, gives tourists a reliable way to eat well, and brings the sector into the digital economy.

3. Who Uses the Platform

The system is built around three clear roles. Each one has its own dashboard and its own set of tools.

The Diner (Client)

A person who wants to eat out. They register with a name, email, phone, and password. They can browse restaurants without even logging in, but to book a table they need an account. They use ChopWise to find a place, view photos and the menu, make a reservation, and pay the booking fee to lock it in.

The Restaurant Owner

The person who manages the restaurant. They sign up, complete a 7-step wizard to set up their restaurant profile (name, location on a map, opening hours, menu, photos, etc.), and wait for an administrator to approve it. Once approved, they receive reservation requests and decide to accept or reject each one within 15 minutes.

The Administrator

The team running ChopWise. Admins review and approve new restaurants before they appear publicly, manage users, suspend or activate accounts, and monitor activity on the platform. The admin role keeps the platform trustworthy by keeping bad actors out.

4. The Complete Set of System Functions

This section lists every function the platform offers today. Each function is grouped by the area of the system it belongs to. If you are evaluating ChopWise, this is the full inventory of what has been built.

4.1 Accounts & Authentication

- **User registration** with name, email, phone, password, and role (Diner or Restaurant Owner). Passwords are encrypted; duplicate emails or phones are rejected with clear errors.
- **Automatic sign-in after registration** — the user does not need to log in again right after creating an account.
- **Email + password login** with friendly error messages ("No account found", "Incorrect password").
- **Profile management** — update name, phone, and profile picture.
- **Password change** with current password verification and strong-password rules.
- **Secure sign-out** available from any dashboard page.
- **Role-based dashboards** — diners, restaurant owners, and admins each see a different home page with different tools.

4.2 Restaurant Onboarding & Profile

- **7-step setup wizard** for restaurant owners: Basic Info, Location, Images, Description, Amenities, Schedule, Review.
- **Drag-to-pin location** on an interactive map (using OpenStreetMap), with the ability to enter coordinates manually.
- **Image uploads** — a banner image and up to 6 gallery images, hosted on Cloudinary's global CDN for fast loading.
- **AI-generated description** — owners can click a button and have OpenAI generate a polished 2-3 sentence description of their restaurant.
- **Amenities selector** — multi-select from 12 predefined options (parking, WiFi, terrace, family-friendly, etc.).
- **Weekly schedule** — set opening and closing times for each day of the week, with the option to mark days as closed.
- **Draft auto-save** — if the owner closes the browser mid-setup, the progress is restored on return.
- **Tabbed profile editor** — after creation, owners can edit any section of their profile at any time.
- **Automatic URL slug** — each restaurant gets a clean web address (e.g., *chopwise.app/restaurants/chez-maurice*).

4.3 Menu Management

- **Add, edit, and delete menu items** with name, description, price in FCFA, category, and an optional photo.
- **Four categories** — Food, Drink, Dessert, Other — each with an item count.
- **Mark items as available or unavailable** with one click, useful when an ingredient runs out.
- **Bulk actions** — mark all items available or unavailable at once.
- **AI-generated item descriptions** — generate a short, appetising description for any dish.
- **Preview mode** — see the menu exactly as a customer will see it.

4.4 Discovery & Search (for Diners)

- **Public restaurant browsing** — anyone can browse restaurants without creating an account.
- **Text search** — search by restaurant name or description.
- **Filters** — by price range (Budget, Moderate, Premium, Luxury), city, and amenities (e.g., only show restaurants with parking and WiFi).
- **Sorting** — newest first, price low-to-high, or price high-to-low.
- **Pagination** — 12 restaurants per page, with shareable URLs (you can send a search result link to a friend).
- **Two views** — a grid of cards, or an interactive map of Cameroon showing every restaurant as a pin.
- **Restaurant detail page** — banner, image gallery with full-screen view, description, amenities, full menu, weekly hours (today highlighted), location map, contact info, and a "Reserve a Table" button.
- **Mobile sticky booking bar** — on phones, a "Reserve" button is always visible at the bottom of the screen.

4.5 Reservation System

- **4-step booking wizard** (Date & Time → Guest Details → Optional Pre-order → Review & Confirm).
- **Smart date and time picker** — only allows future dates (max 30 days ahead), 30-minute intervals, blocks closed days, requires at least 1 hour notice for same-day bookings.
- **Guest count** from 1 to 20 with a +/- counter.
- **Special occasion** — Birthday, Anniversary, Business Meeting, Date Night, etc.
- **Free-text special requests** — allergies, seating preferences, etc.
- **Optional pre-order** — diners can pre-select dishes from the menu so the kitchen is ready when they arrive.
- **Booking fee notice** — the diner is clearly told about the 500 FCFA booking fee before submitting.
- **Live countdown timers** — the diner sees how long the restaurant has to respond; the restaurant sees how long the diner has to pay.
- **Color-coded urgency** — the timer turns red under 2 minutes and pulses under 1 minute.

- **Automatic cancellation** — if the restaurant does not respond within 15 minutes, or the diner does not pay within 30 minutes, the system cancels the reservation cleanly.

4.6 Restaurant-Side Reservation Management

- **Live inbox of incoming requests** — refreshes every 15 seconds when the tab is active.
- **Accept a reservation** — moves it to "awaiting payment" and starts the 30-minute payment window for the diner.
- **Reject with a reason** — Fully booked, Time slot unavailable, Kitchen closed, or a custom reason.
- **Cancel a confirmed reservation** — with a required reason (e.g., emergency closure).
- **Contact the diner** — direct WhatsApp or phone-call link from inside the reservation card.
- **Reservation history and statistics** — see past bookings, cancellations, and counts.

4.7 Payment System

- **Pay Now flow** — diners pay the 500 FCFA booking fee to lock in the table.
- **Payment method selection** — MTN MoMo, Orange Money, or bank card.
- **Phone-number entry** for mobile money.
- **Payment countdown** — the same payment deadline is shown on the checkout page.
- **Receipt page** — green success animation, full booking summary, downloadable/printable receipt with amount, reference number, method, date, and status.
- **Simulated payments today** — payments do not actually charge money yet; a clear yellow banner tells the user it is in test mode.
- **Swap-ready architecture** — switching to a real provider (NotchPay, CinetPay, MTN MoMo API) only requires changing 2 files. The full booking flow does not need to be rebuilt.

4.8 Admin Tools

- **Admin dashboard** with live counts: total users (split by role), total restaurants (approved vs. pending), total reservations, pending approvals.
- **Recent activity feed** — last 10 registered users, last 5 pending restaurants.
- **Restaurant moderation** — tabs for Pending, Approved, Suspended, and All. Search and paginate 20 per page.
- **Restaurant review page** — see the full profile (images, menu, hours, owner contact) before approving.
- **Approve / Reject / Suspend / Activate** a restaurant with one click.
- **User management** — filter by role, search by name or email, suspend or reactivate a user.
- **Safety check** — an admin cannot suspend their own account by mistake.

4.9 Automated Background Jobs

- **Reservation expiry checker** — runs every minute. Automatically cancels reservations where the restaurant did not respond in time, or where the diner did not pay in time.

- **System-attributed cancellations** — these auto-cancellations are tagged as "SYSTEM" with a clear reason, so both sides understand what happened.

4.10 Cross-Cutting Features

- **Mobile-first design** — every page works from a 320px-wide phone screen up to a desktop.
- **Loading skeletons** on every page so the app never looks frozen.
- **Toast notifications** for every action (success, error, info).
- **Error recovery** — a "Try Again" screen catches crashes so the user is never stranded.
- **FCFA throughout** — every price, fee, and total is shown in the local currency.
- **Cameroonian cities supported** — Douala, Yaoundé, Bafoussam, Bamenda, Garoua, Maroua, Limbe, Buea, Kribi, Ngaoundéré.

5. How a Reservation Works, From Start to Finish

This is the single most important flow in the system. Here it is, step by step.

1. A diner browses restaurants in Douala and opens a restaurant page.
2. The diner clicks "Reserve a Table" and fills the 4-step form.
3. The system saves the reservation as `AWAITING_RESPONSE`.
→ A 15-minute timer starts for the restaurant to reply.
4. The restaurant owner sees the request (live, refreshes every 15s).
 - └ Owner `ACCEPTS` → status becomes `PAYMENT_PENDING`.
| → A 30-minute timer starts for the diner to pay.
 - └ Owner `REJECTS` → status becomes `REJECTED` with the reason. End.
 - └ Owner `DOES NOTHING` for 15 min → `SYSTEM` cancels it. End.
5. The diner pays the 500 FCFA booking fee on the checkout page.
 - └ Payment succeeds → status becomes `CONFIRMED`.
 - └ Payment fails / 30-min passes → `SYSTEM` cancels it. End.
 - └ Diner manually cancels → `CANCELLED`. End.
6. Both parties now see the confirmed reservation in their dashboard.
7. Either side can still cancel with a reason if needed.

Why this flow matters: Every step is bounded in time. Restaurants cannot leave diners waiting forever. Diners cannot block a table without committing. The result is a clean, fair, and trustworthy booking experience.

6. The Payment System

6.1 What Happens When a Diner Pays

1. The reservation enters the `PAYMENT_PENDING` state after the restaurant accepts.
2. The diner clicks "Pay Now". The system generates a unique payment reference.
3. The checkout page shows the order summary, available payment methods, and the live countdown timer.
4. The diner selects a method (MTN MoMo, Orange Money, or Card) and enters a phone number.
5. The system processes the payment, marks the reservation as `CONFIRMED`, and records the payment time and method.
6. The diner is redirected to a confirmation page with a receipt they can print or save.

6.2 Why It Is Simulated for Now

Real payment integration in Cameroon (NotchPay, CinetPay, or the MTN MoMo API directly) requires regulatory paperwork and merchant accounts. To allow the platform to launch and gather user feedback first, payments are **simulated** — the full flow works, but no real money moves. A yellow banner on the checkout page makes this clear. The system is built so that switching to real payments later requires changing only two files — the rest of the booking pipeline remains untouched.

7. The Admin Side: Trust & Safety

A platform that connects strangers (diners and restaurants) only works if both sides trust it. The admin tools exist exactly for that.

Admin Action	Why It Exists
Restaurant approval before going public	Stops fake or low-quality listings from appearing to diners.
Suspend a restaurant	If a restaurant repeatedly rejects bookings without reason or behaves badly.
Suspend a user	If a diner abuses the system (e.g., books many tables and never pays).
Live statistics	Helps the operations team monitor platform health day by day.
Self-suspension safeguard	An admin cannot suspend their own account, preventing accidents.

8. Technology Behind the Platform

ChopWise is built on a modern, proven, and affordable technology stack. Everything is serverless, which means hosting costs scale with usage — perfect for a startup.

Layer	Technology	Why It Was Chosen
Web framework	Next.js 16 + React 19	Industry standard, fast page loads, great mobile performance.
Language	TypeScript	Catches bugs before users see them.
Styling	Tailwind CSS + shadcn/ui	Modern, consistent, mobile-first design.
Database	PostgreSQL on Neon	Reliable, serverless, scales automatically.
Authentication	Clerk	Secure session management out of the box.
Images	Cloudinary	Global CDN — fast image loading even on slow networks.
Maps	Leaflet + OpenStreetMap	Free, accurate, no Google Maps fees.
AI	OpenAI GPT-4o-mini	Generates restaurant and menu descriptions.
Hosting	Vercel	Auto-scaling, fast global delivery, simple deployments.

9. What Is Live Today vs. What Is Planned

Feature	Status
User registration & login (Diner, Owner, Admin)	Live
Restaurant onboarding wizard (7 steps)	Live
Menu management with AI descriptions	Live
Public restaurant browsing without login	Live
Search, filters, grid and map views	Live
4-step reservation booking flow	Live
Live countdown timers (response & payment)	Live
Restaurant accept / reject / cancel actions	Live

Feature	Status
Simulated payment system (test mode)	Live
Admin restaurant approval & user management	Live
Automated reservation expiry (every minute)	Live
Image upload to Cloudinary CDN	Live
AI restaurant descriptions (OpenAI)	Live
Real mobile money integration (NotchPay / CinetPay)	Planned
WhatsApp booking notifications	Planned
AI budget-based recommendations ("I have 2,000 FCFA...")	Planned
Reviews & star ratings	Planned
Favorites & wishlist	Planned
Owner analytics dashboard (revenue, trends)	Planned

10. Why ChopWise Matters for Douala

10.1 For the Diner

A young professional in Akwa, an expatriate in Bonapriso, or a family in Bonamoussadi can now plan a Saturday evening on their phone in two minutes. They see what is open, what the menu costs, where it is on the map, and whether the table is theirs — before leaving the house. And when they don't know where to go, they simply tell the ChopWise AI their budget — *"3,000 FCFA, near Akwa, something local"* — and get a curated shortlist back. No more calling five places, no more arriving to find the restaurant fully booked, no more scrolling through Google Maps trying to guess what fits the wallet.

10.2 For the Restaurant Owner

A small restaurant in Deïdo that used to depend on walk-ins now has a free online storefront with photos, menu, and a real booking system. Reservations come in confirmed and paid. The owner spends less time on the phone and more time running the kitchen. Over time, the booking data tells them which dishes sell, which days are slow, and how to grow.

10.3 For the City of Douala

ChopWise digitises an entire sector that was operating on phone calls and word of mouth. It gives tourists a reliable way to find Cameroonian cuisine. It helps small restaurants compete with bigger

ones. It moves transactions into the formal, mobile-money-backed economy. And it creates a foundation that future features — reviews, recommendations, analytics — can build on.

The bigger picture: ChopWise is not just a booking app. It is the first serious attempt to bring Cameroonian restaurants into the digital economy on terms that fit Cameroon — FCFA pricing, mobile money payments, Leaflet maps that work on slow networks, AI that helps small business owners write good descriptions, and a fair booking system that protects both sides.